TALENTRA — DESIGN REVIEW & CORRECTIONS

Milestone 5: B16–B19 | Based on the official docket and current working drafts

Review status: NOT APPROVED FOR FINAL SUBMISSION. This review identifies gaps and decisions; it does not claim team sign-off or executed database tests.

1. Correct milestone mapping

The docket states: **B16** combine entities and columns into the Profile design; **B17** validate the Profile design; **B18** combine entities and columns into the Jobs and Applications design; **B19** validate the Jobs and Applications design. Therefore, B18 is not simply 'validate the ER diagram'.

2. Findings and required corrections

Priority	Finding	Correction required
HIGH	Phase 1 ERD shows organisation and user/account entities, but these are owned by Phase 2 modules in the docket.	Represent them as cross-module dependencies only; document ownership and FK contract, not their internal design.
HIGH	Entity names vary across drafts (e.g. candidate_work_experience vs candidate_experience; interview vs job_interview).	Choose one canonical name for each entity and use it in the entity list, ERD, column inventory, constraints and presentation.
HIGH	Application current_status and status history can become inconsistent.	Decide whether current status is stored and transactionally updated with append-only history, or derived from latest history. Document and enforce one rule.
HIGH	Questionnaire/question entities are shown but the candidate response entity is missing from the ERD.	Add the approved response entity (e.g. application_questionnaire_response) with application/question FKs and uniqueness rules.
HIGH	The docket includes the joined pipeline state, but joining is not shown in the ERD.	Add the approved joining entity/relationship or document exactly how joined state is represented and linked to offer/application.
MEDIUM	job_skill is drawn as a junction but must also exist in the approved entity list and column inventory.	Keep only if approved; define FKs and uniqueness for (job_requirement_id, skill_id).
MEDIUM	Master-data references are placeholders and not fully aligned with the Phase 1 Master Data inventory.	Align lookup entities with the module list; include required lookup references for every fixed enumeration used.
MEDIUM	The ERD omits many entities in the current Milestone 4 inventory, including projects, languages, assessments, course enrolment, shortlist/bookmark, search/view/activity history and referrals.	Either add all approved entities or label this diagram 'core relationship overview', not complete Phase 1 ERD.
MEDIUM	Audit, deletion, nullability, multilingual and sensitive-data rules are not fully represented in the diagram.	Document them in the column/data dictionary and compliance table; the ERD alone is not enough.
MEDIUM	Relationship cardinalities and uniqueness decisions are proposals, not requirements expressly specified by the docket.	Confirm profile-per-account, repeat applications, multiple offers, questionnaire cardinality and joining semantics with the team.
MEDIUM	Some types/nullability in the draft may not meet mandatory domain standards or have written justifications.	Check name length <=150; email <=254 and lowercase where account-identifying; phone E.164; money fixed decimal + currency; all nullable columns justified.

3. Mandatory docket checks still to perform

■ Naming: singular snake_case; id PK; FK uses referenced_table_id; index names idx_table_column; unique constraints uq_table_column.

■ Keys: every entity has a consistent surrogate PK type; natural keys are UNIQUE, not PKs; every FK states ON UPDATE/ON DELETE behaviour.

■ Domains: names up to 150 characters; email up to 254 and lowercase when account-identifying; phone in E.164 with country; money fixed decimal plus currency; timestamps UTC; date ranges use CHECK constraints.

■ Enumerations: fixed values use lookup tables with code, localized display name, active flag and sort order; no free-text enums.

■ Lifecycle/audit: business tables have created_at, created_by, updated_at, updated_by; each entity has explicit soft/hard deletion; soft-delete uses deleted_at and default queries exclude deleted rows; audit is append-only.

■ Nullability/security: every nullable column has written justification; each column is classified Public/Internal/Personal/Sensitive; sensitive data protection is stated.

■ Multilingual: UTF-8 end to end; UI strings by translation key/table; record language for user-entered text when known.

■ Coverage: compare coverage with Naukri, Monster and LinkedIn as required by the docket; record gaps and justification.

4. Correct completion sequence

B16: Combine approved Profile entities, relationships and columns into the Profile design draft.

B17: Validate Profile design against data standards and mandatory/expected coverage; record defects and decisions.

B18: Combine approved Jobs & Applications entities, relationships and columns into its design draft, including questionnaire responses and the joined pipeline outcome if approved.

B19: Validate the Jobs & Applications design: relationships, types, nullability, lookups, audit/lifecycle, constraints and cross-module boundaries.

5. Correct conclusion

The current ERD is useful as a **core relationship sketch**, but it is not a complete or validated Phase 1 ERD. Do not submit it as the final signed design until names are reconciled, omitted entities are handled, cross-module ownership is labelled, questionnaire responses and joining are modelled, and the companion column/data dictionary and rule-compliance table are reviewed.

Source basis: Talentra Database Design Hackathon Docket, sections 4–7, plus current working Milestones 1–4 drafts. Choices not explicitly stated by the docket are flagged for team confirmation rather than treated as source requirements.